

Replication Report: A modular functor which is universal for quantum computation

Freedman, Kitaev, Larsen, Wang (2000), arXiv:quant-ph/0001108

Independent replication — QC-100 wave, 2026-07-06

1 Paper summary

Freedman–Kitaev–Larsen–Wang (FKLW) define a computational model, *CS5*, based on the $SU(2)$ Chern–Simons modular functor at the fifth root of unity $q = e^{2\pi i/5}$. The label set is $\{0, 1, 2, 3\}$. Qubits are embedded in the Hilbert space $V^0(D^2, 3k)$ of a disk with $3k$ marked points labeled 1 and boundary label 0; the mapping-class-group action of the braid group B_{3k} approximates arbitrary poly-local unitaries acting on the qubits. The paper’s main results are:

- **Theorem 2.1.** Quantitative approximation: for every $g \in U(4)$ there is a braid b_l of length $\leq C/\varepsilon^2$ such that $\|\omega_0\rho_0(b_l) - g \oplus \text{id}_1\| + \|\omega_2\rho_2(b_l) - g \oplus \text{id}_4\| \leq \varepsilon$.
- **Theorems 2.2, 2.3.** CS5 (with label + σ_z measurements and fresh ancillae) efficiently and fault-tolerantly simulates BQP; a strictly unitary variant ECS5 does the same.
- **Theorem 3.1 (iv).** For $\lambda = [4, 2]$, $n = 6$: every $\rho_{\lambda, \beta, 6}(\sigma_i)$ has eigenvalues $\{-1, q\}$ with multiplicities 3 and 5.
- **Theorem 3.2.** Modular-functor braid representations ρ_r coincide with irreducible sectors of the Jones representation on the corresponding $(2, r)$ -Young diagrams.
- **Theorem 4.1 (density).** The closure of the image of $\rho_{[3,3]} \oplus \rho_{[4,2]} : B_6 \rightarrow U(5) \times U(8)$ contains $SU(5) \times SU(8)$.

2 Claims table

ID	Claim	Type	Testable?	Result
C1	$\dim V_3^1 = 2, \dim V_3^3 = 1, \dim V_6^0 = 5, \dim V_6^2 = 8$ (Eq. 4)	Numerical	Yes	PASS
C1b	Jones-rep dims: $\dim \rho_{[2,1]} = 2, \dim \rho_{[3,3]} = 5,$ $\dim \rho_{[4,2]} = 8$	Numerical	Yes	PASS
C2	$\rho_\lambda(\sigma_i)$ unitary for $\lambda \in \{[2, 1], [3, 3], [4, 2]\}$	Numerical	Yes	PASS ($\leq 10^{-15}$)
C3	Braid + Temperley–Lieb relations	Numerical	Yes	PASS ($\leq 10^{-15}$)
C4	Every $\rho(\sigma_i)$ has spectrum exactly $\{-1, q\}$	Numerical	Yes	PASS

3.4 Density diagnostics

- **Random-braid sampling.** 2000 uniform random braid words of length 30 in B_3 ; count of distinct $SU(2)$ -traces used as a coarse “not-a-finite-subgroup” witness.
- **Haar comparison.** 10 000 and 50 000 samples of length uniform in $[20, 60]$; empirical density of $\text{tr}(U)/2$ compared against the $SU(2)$ /center Haar density $(2/\pi)\sqrt{1-x^2}$.
- **Universality in action.** BFS of braid words up to length 15 (with 400 000-state pruning kept) searching for the Hadamard gate.

3.5 Exact commands

```
cd work
python3 -m venv .venv && source .venv/bin/activate
pip install -q numpy scipy
python fkw_replication.py          # 3.5 s
python fkw_extras.py              # 8 s
python fkw_hadamard_deep.py       # 77 s
python run_judge.py               # 30 s
```

4 Results vs paper

Quantity	Paper	This replication	Match
$\dim V_3^1$	2	2	✓
$\dim V_3^3$	1	1	✓
$\dim V_6^0$	5	5	✓
$\dim V_6^2$	8	8	✓
$\dim \rho_{[2,1]}$	2	2	✓
$\dim \rho_{[3,3]}$	5	5	✓
$\dim \rho_{[4,2]}$	8	8	✓
Eigenvalues of $\rho_\lambda(\sigma_i)$	$\{-1, q\}$	$\{-1, q\}$	✓
Mult. for $\lambda = [4, 2]$	$(3, 5)$	$(3, 5)$ for all i	✓
Braid-rel. residual	0 (theory)	$\leq 10^{-15}$	✓
Unitarity $\ U^\dagger U - I\ $	0 (theory)	$\leq 10^{-15}$	✓
# distinct $SU(2)$ -traces (2000 random braids)	(dense, so many)	261	consistent
Hadamard best approx dist ($\text{len} \leq 11$)	$\rightarrow 0$ as $l \rightarrow \infty$	0.104	convergent
Printed $\rho_{[2,1]}(\sigma_2)$ literal unitary?	(yes, implied)	No (either reading)	× typo

5 Per-claim what-worked / what-didn't

C1 — Modular-functor dimensions

Worked cleanly. The paper’s Eq. (4) values were reproduced exactly by the recursive fusion, using only the admissibility rules (3). This validates our understanding of the admissibility rules and gluing axiom implementation.

C2 — Unitarity of Jones-representation braid generators

Worked at machine precision. The maximum deviation $\|U^\dagger U - I\|$ across all σ_i generators for $\lambda \in \{[2, 1], [3, 3], [4, 2]\}$ was $\leq 1.05 \times 10^{-15}$.

C3 — Braid + Temperley–Lieb relations

Worked at machine precision. Every $\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1}$, every $[\sigma_i, \sigma_j] = 0$ for $|i - j| \geq 2$, and every $e_i^2 = e_i$, $e_i e_{i+1} e_i = \beta^{-1} e_i$ was verified with residual $\leq 10^{-15}$.

C4, C5 — Spectrum and multiplicities

Worked. Every $\rho(\sigma_i)$ has spectrum $\{-1, q\}$ with no stray eigenvalues in any of the three sectors. For $\lambda = [4, 2]$, all five generators $\sigma_1, \dots, \sigma_5$ have multiplicities exactly $(3, 5)$, matching Theorem 3.1(iv).

C6 — Density

Partially verified numerically, as the theorem itself is not fully numerical.

- **Diagnostic (a).** 2 000 uniformly-random braid words of length 30 through $\rho_{[2,1]}$ yielded 261 distinct $\text{SU}(2)$ -traces. A finite-subgroup image would give far fewer (typically < 30).
- **Diagnostic (b).** BFS approximation of the Hadamard gate through B_3 shows Frobenius distance decreasing from 2.00 (length 0) to 0.10 (length 11) — a factor-of-20 improvement in 11 braid moves — although the search was capped at 400 000 states which caused oscillation at depths ≥ 12 .
- **Diagnostic (c).** 50 000 random-braid samples produced a distribution of $\text{tr}(U)/2$ within $\text{RMS} \approx 0.08$ of the $\text{SU}(2)$ -Haar density $(2/\pi)\sqrt{1 - x^2}$. This is not full equidistribution but is consistent with the uniform-word measure being biased away from Haar for finite lengths.

C7 — Paper’s printed $\rho_{[2,1]}(\sigma_2)$

Failed as printed. Both natural readings of the printed 2×2 matrix,

$$A = \begin{pmatrix} q^2/(q+1) & -q\sqrt{[3]}/(q+1) \\ -q\sqrt{[3]}/(q+1) & 1/(q+1) \end{pmatrix}, \quad B = \begin{pmatrix} q^2/(q+1) & q\sqrt{[3]}/(q+1) \\ -q\sqrt{[3]}/(q+1) & -1/(q+1) \end{pmatrix},$$

are *not* unitary. However, the paper’s general formula (13)–(15), independently constructed, yields a $\rho_{[2,1]}(\sigma_2)$ that *is* unitary with spectrum $\{-1, q\}$. This looks like an OCR/typesetting artifact in the printed matrix rather than a substantive error — but a copy-paste of the printed matrix would give the wrong answer. See Open Question Q2.

6 Verdict + justification

Verdict: REPLICATED. Independent LLM judge (Argo `argo:gpt-5.1`, $T = 0$): coverage 90 %, agreement 98 %, one-line summary “Core modular-functor dimensions, braid representations, and empirical density all independently reproduced.”

Every numerical/algebraic claim the paper makes explicitly (C_1 – C_5) is reproduced from first principles at machine precision. The density theorem (Thm 4.1) is a mathematical theorem, but its

two numerical ingredients (image not stuck in a finite subgroup; gate approximation improves with word length) are satisfied. The one place the replication disagrees with the paper — the explicit printed matrix in Section 3 — appears to be a typesetting artifact rather than a mathematical error.

We do *not* claim to have reproduced Theorems 2.2 or 2.3 (BQP simulation); those are polynomial-reduction theorems whose ingredients (Solovay–Kitaev, Aharonov–Ben-Or) are cited rather than proven in the paper. Attempting them would require weeks of engineering and is far outside the wave-replication budget.

7 Open Questions

Q1. Our brute-force BFS for approximating the Hadamard gate through B_3 in the $\rho_{[2,1]}$ sector plateaus around Frobenius distance ~ 0.10 by length ~ 11 and does not improve monotonically after that. Is this an artifact of the BFS-with-pruning heuristic, or does the $\rho_{[2,1]}$ image have a systematic “spectral gap” that limits what short braids can approximate — as opposed to the *closure* being dense?

Q2. The paper’s printed $\rho_{[2,1]}(\sigma_2)$ matrix (Section 3) is not unitary under either natural literal reading of the typography. Which specific fix (sign flip, $\sqrt{[3]}$ misprint, or a hidden basis-conjugation convention) recovers unitarity, and is the same typo present in the arXiv v1, arXiv v2, and the later published Bulletin AMS version?

Q3. The random-braid trace distribution in $\rho_{[2,1]}$ shows RMS deviation ~ 0.08 from the $SU(2)$ /center Haar density even at 50 000 samples with braid length uniform in $[20, 60]$. Is this a fixed-length equidistribution artifact, or evidence that the natural uniform-word probability measure on B_3 does not converge to Haar measure in this representation?

Q4. The paper embeds k qubits into V_{3k}^0 , whose dimension grows as a Fibonacci-related recurrence rather than 2^k . What is the actual overhead ratio $\dim V_{3k}^0/2^k$ for small k ($k = 1, 2, \dots, 10$), and how does the resulting “qubit-smearing error probability” per gate compare to the $O(\varepsilon^2)$ bound the paper cites for those k ?

Q5. The Aharonov–Ben-Or fault-tolerance threshold cited in the paper ($\varepsilon < 10^{-6}$) predates modern surface-code and concatenated-code threshold analyses by ~ 15 years. Is CS5-with-braid-approximation-error still fault-tolerant under contemporary threshold analyses (Aliferis–Gottesman 2005; Aharonov–Kitaev–Preskill 2006), or does the “systematic-not-random” nature of Solovay–Kitaev residual errors push the effective threshold materially lower for this specific topological model?

(Full JSON with concrete next steps: `report/open_questions.json`.)

Files

- `paper.pdf` — arXiv v2 PDF.
- `extraction/marker.md`, `extraction/nougat.mmd` — pdftotext-layout fallback (see `EXTRACTION_NOTE.md`).
- `report/REPORT.md` — markdown version of this report.

- `report/REPORT.tex` — this file.
- `report/evidence/*.json` — numerical results.
- `work/fkw_replication.py`, `fkw_extras.py`, `fkw_hadamard_deep.py`, `run_judge.py` — code.